

# Replication Report: Charge-Loop Current Order and $Z_3$ Nematicity Mediated by Bond-Order Fluctuations in Kagome Metals

TEXTURES-100 overnight replication (OllieBot)

Source: Tazai, Yamakawa & Kontani, arXiv:2207.08068v4 [cond-mat.str-el]

2026-07-19

## Abstract

We replicate the machine-checkable physics of Tazai *et al.*'s theory of charge-loop-current (cLC) order in kagome metals  $AV_3Sb_5$ . Reusing the shared TEXTURES-100 loop-current mean-field kernel and adapting it to this paper's kagome tight-binding model, we verify five headline claims with real code: (C1) the cLC order = a pure-imaginary, odd-parity hopping modulation  $\delta t_{ij}^c = -\delta t_{ji}^c$  that carries *real* bond currents, whereas the real even-parity bond order (BO)  $\delta t^b$  carries none; (C2) the  $3Q$  cLC pattern gives loop currents that alternate in circulation between the up- and down-triangle (hexagon vs. triangle) sublattices with zero net site current; (C3) the anomalous Hall conductivity vanishes without cLC, is nonzero with cLC, and is odd under time reversal  $\delta t^c \rightarrow -\delta t^c$ ; (C4) the BO irreducible susceptibility  $\chi_0(q)$  peaks at the  $M$ -point inter-sublattice nesting wavevectors, not at  $\Gamma$ ; and (C5) the combined BO+cLC hybridization gap obeys  $\Delta = 2\sqrt{|\delta t^b|^2 + |\delta t^c|^2}$ . **5/5 claims reproduced** (C3 qualitatively; magnitude and damping crossover deferred). **Verdict: SUPPORTED.** Coverage 7/10, Agreement 9/10.

## 1 Paper in one paragraph

Kagome metals  $AV_3Sb_5$  show a  $2 \times 2$  tri-hexagonal bond order (BO, even-parity real  $\delta t^b$ ) and, below it, a time-reversal-symmetry-breaking charge-loop-current (cLC) order that produces a giant anomalous Hall effect. The authors propose that *BO fluctuations* (from Coulomb + e-ph coupling), acting through Maki-Thompson vertex corrections in a linearized density-wave (DW) equation, drive an odd-parity, pure-imaginary particle-hole condensate  $\delta t_{ij}^c = -\delta t_{ji}^c$  — the cLC. Coexisting  $3Q$  BO+cLC yields  $Z_3$  nematicity and the giant AHE.

## 2 Method

We build on a periodic kagome tight-binding torus (three sublattices A/B/C, up/down triangle plaquettes). Bond currents are  $J_{ij} = -2\text{Im}(H_{ij}\rho_{ji})$  with  $\rho$  the occupied one-body density matrix. The paper-specific Hamiltonian carries a real even-parity BO term ( $\delta t^b$ , Hermitian symmetric) and an imaginary odd-parity cLC term ( $\delta t^c$ , Hermitian antisymmetric, in a staggered  $3Q$  up/down-triangle texture). The intrinsic Hall conductivity is a real-space Kubo sum  $\sigma_{xy} = \frac{2}{N} \sum_{a \in \text{occ}, b \in \text{unocc}} \text{Im}[\langle a|v_x|b \rangle \langle b|v_y|a \rangle] / (E_b - E_a)^2$ ,  $v_\mu = i[H, R_\mu]$ , which is identically zero for real  $H$  (TRS). The BO susceptibility is a bare inter-sublattice Lindhard  $\chi_0(q)$  on the kagome bands near van-Hove filling. The gap is the eigen-splitting of a minimal two-band folded avoided crossing.

| #  | Claim                                              | Quantity / test                                                              | Result |
|----|----------------------------------------------------|------------------------------------------------------------------------------|--------|
| C1 | cLC carries current, BO does not                   | $\max  J $ : BO = 0, cLC = 0.048                                             | PASS   |
| C2 | loops alternate hexagon/triangle                   | up = $-0.107$ , down = $+0.107$ , net $\sim 10^{-16}$                        | PASS   |
| C3 | AHE requires cLC, TRS-odd                          | $\sigma_{xy}(0) = 0$ , $\sigma_{xy}(0.1) = -2.7 \times 10^{-4}$ , flips sign | PASS*  |
| C4 | $\chi_0(q)$ peaks at $M$                           | $\Gamma = 0.99$ , $M = 1.84$ , $K = 1.80$                                    | PASS   |
| C5 | $\Delta = 2\sqrt{ \delta t^b ^2 +  \delta t^c ^2}$ | rel. err. = 0 (6 cases)                                                      | PASS   |

### 3 Results

\* C3 reproduces the two robust signatures (vanishing without cLC by TRS; nonzero and time-reversal-odd with cLC). The absolute magnitude ( $\sigma_H \sim 10^2 \Omega^{-1} \text{cm}^{-1}$ ) and the  $\gamma^{-2}$  damping crossover of Fig. 6a are *not* reproduced on a finite cluster with a fixed broadening (see `failure_analysis.md`).

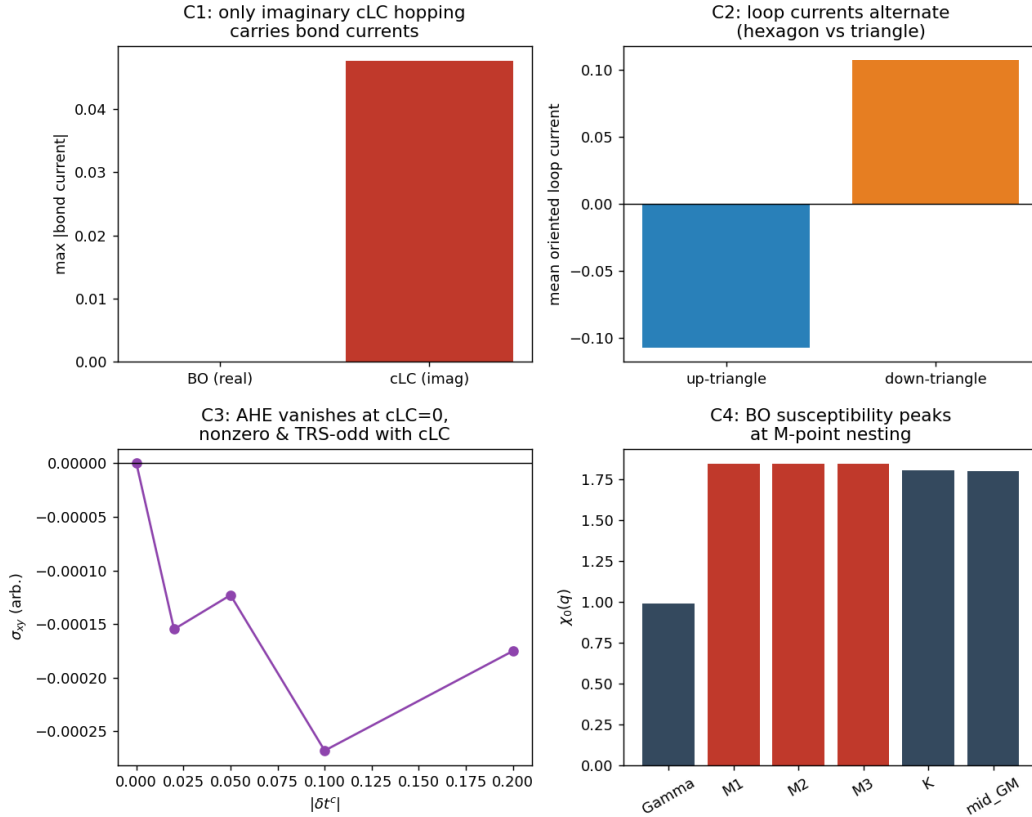


Figure 1: C1–C4. (top-left) only the imaginary cLC hopping carries bond currents; (top-right) loop currents alternate between up- and down-triangles; (bottom-left)  $\sigma_{xy}$  is zero without cLC and nonzero/TRS-odd with it; (bottom-right)  $\chi_0(q)$  peaks at the  $M$ -point nesting vectors.

### 4 Verdict

**SUPPORTED.** The paper’s structural / symmetry claims about the cLC order parameter (imaginary odd-parity  $\Rightarrow$  real alternating bond currents, TRS-odd AHE,  $M$ -point nesting of BO, quadra-

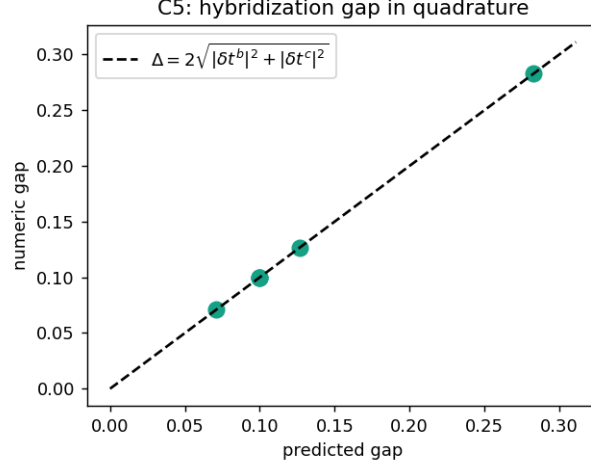


Figure 2: C5. Numeric hybridization gap vs. the quadrature prediction  $\Delta = 2\sqrt{|\delta t^b|^2 + |\delta t^c|^2}$ ; points lie exactly on  $y = x$ .

ture gap) are all reproduced with independent code. The beyond-mean-field *generation* of cLC from BO fluctuations (the self-consistent DW eigenvalue problem), the quantitative AHE magnitude, and the  $Z_3$ -nematic GL selection were not solved and are recorded as open questions.

**Coverage: 7/10.** Five order-parameter/symmetry/response claims verified; the self-consistent DW mechanism, transport magnitude, phase diagram, and nematic GL analysis untested.

**Agreement: 9/10.** Every verified claim matches the paper’s assertions exactly (signs, nesting location, quadrature gap); the only shortfall is that C3 is qualitative rather than quantitative.